Data First vs Integration First:
Why Digital Government Integration Fails Without Data Governance

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Abstract
This paper analyzes the limitations of the integration-first approach to digital government, using the case of Kyrgyzstan and Central Asian countries. Despite significant investments in integration platforms, the expected outcomes - end-to-end digital services and effective data exchange - often remain limited.
The study demonstrates that the key underlying issue is the lack of a mature data governance framework, including data inventory, standardization, and clear assignment of responsibilities. Based on a qualitative case study, the paper proposes a conceptual model identifying two levels of data-related constraints: the "unknown data landscape" and "semantic fragmentation."
As a practical contribution, the paper introduces the Data-First Digital Government model, outlining a sequence for transitioning from fragmented systems to a coherent data environment.
The findings can inform public policy design in digital transformation and support the development of government data architectures.

Keywords:
digital government, data governance, interoperability, public sector, GovTech, Central Asia, data architecture, e-government

Research Questions
This study aims to address the following questions:
1. Why does the integration-first approach to digital government often fail to deliver expected results?
2. What data-related structural constraints underlie these challenges?
3. How can a data-first approach help overcome these limitations?

Methodology
This paper employs a qualitative case study approach, focusing on the development of digital government infrastructure in the Kyrgyz Republic. The analysis draws on a combination of sources, including regulatory documents, analytical materials, and the authors' practical experience in digital transformation projects.
The study is interpretative in nature and aims to identify structural constraints related to data governance that are not always captured through quantitative performance assessments.
To broaden the analytical context, a comparative review of Central Asian countries is included, based on publicly available sources and current policy directions in digital development.

Contribution
This study contributes to the development of digital government approaches by proposing:
(1) a distinction between system interoperability and data interoperability;
(2) a two-level model of data-related constraints;
(3) a practical model for transitioning to a data-first approach based on a regional case.

Introduction
Over the past 15 years, governments around the world have invested significant resources in integration platforms for public sector information systems. However, in

many cases, the expected outcomes - seamless digital services and efficient data exchange - have not been fully achieved.
The key reason for this paradox lies in the fact that governments attempt to integrate systems without a clear understanding of what data they possess and how compatible those data are.
Despite a substantial body of research on interoperability and digital government, existing approaches largely focus on technological and architectural aspects of integration. At the same time, insufficient attention is given to the structural characteristics of data as a critical factor determining the effectiveness of digital transformation.
This study aims to address this gap by analyzing digital government through the lens of data governance, focusing on limitations related to data structure, consistency, and accountability.

In this context, integration should not be viewed as the starting point of digital transformation, but rather as a function of data governance maturity. This implies that the ability of systems to interact depends not only on technical infrastructure, but also on the quality, consistency, and manageability of the underlying data.
System interoperability does not necessarily imply data interoperability.

Global Context

Over the past two decades, approaches to digital transformation in the public sector have undergone significant evolution. Early e-government initiatives primarily focused on automating individual government functions and moving services online. In this model, information systems of individual agencies were the main objects of digitalization, and the primary goal was to improve internal efficiency.
However, as the number of such systems grew and inter-agency interactions became more complex, it became evident that fragmented automation could not deliver systemic impact. This led to a shift toward the concept of digital government, where the focus moves from individual systems to the overall architecture of public administration.
According to frameworks developed by the World Bank[1] and OECD[2], modern digital government is increasingly understood as a data-driven system, where data is treated as a strategic asset. In this paradigm, digital services, platforms, and integration solutions are derived from how governments manage their data - its structure, quality, accessibility, and interoperability.
This shift fundamentally changes the logic of digital transformation. While in the e-government paradigm data was often treated as a byproduct of information systems, in digital government it becomes the central element around which architecture, processes, and services are built.
In this context, a fundamental distinction emerges between two approaches to digitalization: integration-first and data-first.
The integration-first approach assumes that building data exchange infrastructure - such as integration platforms, service buses, and APIs - will enable interaction between government systems. In contrast, the data-first approach assumes that without first establishing a coherent data environment - including data inventory,

[1] World Bank, Digital Government (2020)
[2] OECD Digital Government Policy Framework

standardization, and clear ownership - any integration efforts will remain limited in effectiveness.
Recent practice shows that underestimating the role of data as the foundation of digital government is one of the key reasons why large-scale investments in integration infrastructure often fail to deliver the expected outcomes.

Typical Digitalization Path and Systemic Error

Digital transformation in the public sector across different countries follows a remarkably similar trajectory, regardless of economic development or institutional maturity.
At the first stage, governments invest in automating individual agencies. Information systems are developed, internal processes are digitized, and initial electronic services are introduced. These systems are typically created independently and reflect the organizational structure of government, with each system serving a specific agency and its functions.
At the second stage, the need for inter-agency interaction emerges. Citizens and businesses expect seamless services that require data exchange between multiple government bodies. In response, governments begin to develop integration mechanisms - initially point-to-point integrations, followed by centralized data exchange platforms such as service buses and API gateways.
At this stage, a persistent assumption takes hold: that fragmentation can be solved through integration technologies. The logic appears straightforward - if systems do not interact, build infrastructure to enable their interaction.
However, after launching integration platforms, many countries encounter an unexpected outcome. Despite having the technical capability to exchange data, actual integration either progresses very slowly or fails to produce meaningful improvements in public services.
What emerges can be described as a systemic data crisis, which becomes a key constraint on further digital transformation.
This crisis manifests in several ways. First, there is no comprehensive understanding of what data exists within the government and where it resides. Second, even when technical channels for data exchange exist, the data itself is often incompatible - different classifications, identifiers, and data structures are used. Third, responsibility for data is unclear - there is no well-defined ownership or accountability for data quality.
As a result, integration infrastructure begins to serve a different purpose than originally intended. Instead of enabling efficient data exchange, it becomes a mechanism for transferring fragmented and inconsistent information between systems, limiting the ability to create truly seamless and proactive public services.

Thus, the core issue is not the lack of integration technologies, but the fact that integration is treated as the starting point of digitalization rather than its outcome.
This gap between technical readiness for data exchange and the absence of a mature data governance environment lies at the heart of many unsuccessful or limited integration initiatives in the public sector.

Case Insight: Kyrgyzstan

As a foundation for inter-agency interaction, Kyrgyzstan implemented the Interagency Electronic Interaction System “Tunduk,” built on the X-Road architecture originally developed in Estonia. This technology is widely recognized as a core component of

Estonia’s digital government model and enables secure, decentralized data exchange between information systems.
Architecturally, the system follows a classic integration bus model: each government agency independently develops and publishes services through which other participants can request data. This approach ensures flexibility and scalability while eliminating the need for centralized data storage.
In terms of scale, the system demonstrates a high level of development. Approximately 380 participants - including government agencies and private sector entities - are connected to the platform. More than 2,100 services have been implemented to support inter-agency data exchange. The volume of transactions has reached significant levels, with around 3.9 billion operations in the G2G segment and approximately 1.5 billion in the G2B segment. Nearly 400 data exchange processes are currently in operation.
However, this scale of integration has not led to a corresponding level of data manageability.
These indicators suggest that, at the level of technological infrastructure, Kyrgyzstan has replicated one of the most advanced global approaches to inter-agency interaction. Yet this technological advancement has not been accompanied by a comparable level of data governance maturity.

As the system has scaled, a set of limitations has emerged that are not related to the underlying technology.
First, there is no comprehensive view of government data. Despite the presence of a developed data exchange infrastructure, no centralized data catalog has been established to systematically identify what data exists, where it resides, and which institutions are responsible for it. In practice, the only formalized registry is the list of integration services, which reflects implemented exchange scenarios rather than the structure of data itself. As a result, knowledge about available data remains fragmented and is often discovered only during the development of specific integrations.
Second, there is a lack of semantic consistency in data. Government information systems have been developed independently, within the scope of agency-specific tasks and without unified requirements for data models. This has led to the use of different classifications, reference data, and identifiers. As a result, even when technical data exchange is possible, additional interpretation and transformation are required, reducing efficiency and complicating the development of scalable digital services.
Third, the absence of a formalized data governance system further exacerbates the problem. Roles such as data owners and sources of truth are not sufficiently defined, leading to risks of duplication, inconsistencies, and reduced trust in data when used across agencies.

Thus, the case of Kyrgyzstan illustrates a critical structural gap: despite the use of an advanced integration technology proven in other countries, there is no equivalent level of maturity in data governance.
This gap highlights a key insight: technological interoperability does not equal data interoperability. Without a coherent system of classifications, unified data models, and transparent data governance, even the most advanced integration platforms tend to scale fragmentation rather than connectivity.

At the same time, the existence of a functioning integration platform creates favorable conditions for transitioning to a data-first model, as the foundational data exchange infrastructure is already in place.

Problem Analysis: From “Unknown Data” to Semantic Fragmentation

The analysis of digital transformation practices, including the case of Kyrgyzstan, allows for the identification of two interrelated but conceptually distinct levels of problems that hinder effective integration of government systems. These levels provide a structural explanation for why the presence of integration infrastructure does not automatically lead to digital government outcomes.

1. The Problem of the Unknown Data Landscape

The first level relates to the lack of a comprehensive understanding of government data.
In the traditional model of managing information systems, data is “locked” within individual systems. Each agency owns its system, and knowledge about data is effectively embedded within it. As a result, no unified view emerges at the level of the state regarding what data exists, where it resides, and how it can be used.

This leads to several systemic consequences.
First, integration cannot be planned based on data. Data exchange is initiated not because specific datasets are known to exist, but because a concrete need arises within a particular project or service. Integration thus becomes reactive rather than proactive.

Second, dependence on institutional memory and informal knowledge increases. Information about available data often resides with specific individuals or teams and is not formalized in shared registries or data catalogs.

Third, responsibility for data remains unclear. Without clearly defined data owners and sources of truth, it is difficult to ensure data quality, consistency, and reliability in cross-agency use.
Thus, the problem of the “unknown data landscape” makes data itself difficult to manage as an asset, even when a developed exchange infrastructure exists.

2. The Problem of Semantic Fragmentation

The second level of problems arises even when data is technically accessible.
Semantic fragmentation refers to the lack of consistency in how data is defined, structured, and interpreted across different systems. While technical interoperability enables data exchange, semantic interoperability determines whether that data can be correctly understood and effectively used.
In practice, this manifests in the use of:

- different classifications and reference data for the same entities;
- incompatible identifiers;
- divergent data models and storage structures.

As a result, data that is conceptually identical may be represented differently across systems, requiring additional efforts for mapping and transformation. These processes

are typically implemented within integration services, leading to increased architectural complexity and reduced scalability.
Moreover, in the absence of common standards, such transformations are often duplicated across multiple integrations, creating hidden complexity and increasing maintenance costs.
This ultimately results in higher integration costs, duplicated solutions, and limited scalability of digital services.

3. Interrelation of the Two Levels
A key characteristic of these problems is that they reinforce each other.
The lack of visibility into data (the first level) makes it difficult to identify and systematically address semantic inconsistencies (the second level). In turn, a high degree of semantic fragmentation reduces the value of even those data assets that are known to exist.
Together, these factors create a situation in which integration infrastructure operates under conditions of uncertainty and inconsistency, limiting its ability to serve as a foundation for digital services.

4. Key Insight
Distinguishing these two levels provides a new perspective on the challenges of digital government.
The limitations of integration are not primarily technological, but rather stem from the absence of a mature data governance environment.
Until governments establish a transparent and coherent data landscape, scaling integration will inevitably lead to increased complexity and reduced effectiveness.

Global Experience: From Integration to Data Governance

International experience in public sector digital transformation over recent years demonstrates a gradual but consistent shift from technology-driven approaches toward models centered on data governance.
Early initiatives in inter-agency interaction across many countries focused primarily on building integration platforms and data exchange infrastructure. However, accumulated experience has shown that even the most advanced technological solutions do not guarantee interoperability at the data level.
In response, more comprehensive approaches have emerged at both international and national levels, incorporating not only technical, but also semantic and organizational dimensions.
Frameworks promoted by the World Bank and OECD emphasize the transition toward a data-driven government model, where data is treated as a strategic asset requiring systematic governance. Within this framework, key elements include not only exchange platforms, but also data governance mechanisms such as defining data ownership, standardization, quality assurance, and the establishment of data catalogs.
A similar approach is reflected in the European Interoperability Framework[3], which explicitly highlights the need to ensure interoperability across multiple layers: legal, organizational, semantic, and technical. Importantly, the technical layer is viewed as just one component rather than the foundation of the entire system.

[3] European Interoperability Framework

The case of Estonia - often considered a benchmark for digital government - is also illustrative. While the X-Road[4] platform is widely recognized for enabling inter-agency data exchange, its effectiveness is not solely due to technology. It is supported by a well-structured data environment, including core registries, unified identifiers, and harmonized classifications, which ensure semantic consistency of data.

Thus, even in the most successful cases, integration infrastructure functions not as a starting point, but as part of a broader data governance ecosystem.
Further evidence of this shift can be seen in the growing adoption of the Government as a Platform[5] (GaaP) concept, in which the state is viewed as a platform providing standardized data and services for reuse. In this model, core registries, shared services, and data standards are central components, while integration serves as an enabling layer.
Overall, international experience suggests that sustainable digital transformation outcomes are achieved not by scaling integration solutions alone, but by establishing a coherent data governance environment within which integration becomes a natural and relatively straightforward process.
This reinforces the conclusion that data-level interoperability is a necessary condition for effective digital transformation.

Regional Context: Central Asia as a Shared Trajectory

The experience of Central Asian countries reveals a broadly similar trajectory of digital government development, despite differences in maturity levels and scale.
Kazakhstan represents the most advanced example in the region. The country ranks highly in international indices, including the UN E-Government Development Index, and has achieved a significant level of digitalization of public services, with over 90% of services available online and millions of active users.
This demonstrates that large-scale digitalization and the development of integration infrastructure can significantly improve service accessibility and user engagement.
At the same time, Kazakhstan's current digital agenda is increasingly shifting toward data governance and the use of data for decision-making and the deployment of artificial intelligence. This reflects a broader transition from e-government to data-driven government, where data becomes the key resource.
In less mature digital environments - such as Kyrgyzstan, Uzbekistan, and Tajikistan - this transition is at an earlier stage. However, the structural challenges are similar: the lack of a comprehensive data landscape and the absence of semantic consistency.
In Uzbekistan, recent years have seen rapid growth in digital public services and e-government infrastructure, including the development of a unified public services portal, expanded inter-agency interaction, and the introduction of digital identity solutions. At the same time, the policy agenda is increasingly shifting toward integrating government registries and improving data governance.
In Tajikistan, digitalization remains at an earlier stage, with significant involvement of international organizations. Key challenges are evident even at a foundational level, including limited standardization, fragmented systems, and the absence of unified data.
This suggests that data governance challenges are not a consequence of integration, but a fundamental factor shaping the entire trajectory of digital transformation.

[4] Estonia X-Road
[5] Government as a Platform (UK / Tim O'Reilly)

As a result, a consistent regional pattern emerges: integration infrastructure tends to develop faster than data governance systems.

To systematize differences in the maturity of digital government across the region, a comparative overview is presented below.

Country	Level of Integration	Data governance	Main Challenge
Kazakhstan	High	Growing	data utilization
Kyrgyzstan	High	Low	data inventory
Uzbekistan	Medium	Emerging	integration → data
Tajikistan	Low	Low	Basic standardization

Table 1. Comparative maturity of digital government in Central Asian countries (author's assessment)

The assessment is qualitative and based on analysis of open sources and practical experience with digital initiatives.

Data-First Digital Government: A Practical Model

Analysis of digital transformation practices and international experience shows that effective transformation requires a shift toward a data-first approach. Unlike the integration-first model, this approach assumes that sustainable inter-agency interaction and digital services are derived from a mature data environment.
The proposed model can be described as a sequence of interconnected stages.

1. Data Inventory
The first step is to establish a comprehensive understanding of government data.
In practice, this involves creating a data catalog that captures:

- what data exists;
- where it is stored;
- which institutions are responsible for it;
- in what format and with what level of quality it is available.

A key feature of this stage is the shift in focus from systems to data: the unit of analysis becomes not the information system itself, but the datasets it contains.
Without this step, integration initiatives remain reactive, as there is no clear understanding of what data can and should be used.

2. Data Governance
The next stage involves institutionalizing data governance.
Key elements include:

- defining data owners;
- assigning responsibility for data quality and accuracy;
- establishing rules for access and data exchange;
- implementing data change management processes.

At this stage, data becomes a formally governed asset at the state level, rather than a byproduct of individual systems.

3. Standardization
Once data is identified and responsibilities are defined, the next step is to ensure consistency.
This stage includes:

- developing and implementing unified classifications and reference data;

- standardizing identifiers (for individuals, legal entities, and objects);
- harmonizing data models and exchange formats.

It is at this level that the problem of semantic fragmentation is addressed, removing one of the key barriers to effective integration.

4. Core Registries

The next step is to define “sources of truth” for key data domains.
Typically, these include:

- population registry;
- business registry;
- address register;
- registries of assets (real estate, vehicles, etc.).

For each domain, a single source of truth is established, reducing duplication and increasing trust in data across systems.

5. Interoperability Layer

Only after establishing a coherent data environment does integration infrastructure begin to fulfill its intended function.
At this stage:

- data exchange platforms (e.g., X-Road-type solutions) are implemented or scaled;
- access interfaces (APIs) are standardized;
- secure and controlled data exchange is enabled.

Importantly, in this model, integration is not the starting point but a logical continuation of previous stages.

6. Service Layer

The final stage involves building digital services based on already available and consistent data.
This enables:

- the creation of seamless services without repeated data collection;
- the development of proactive services;
- the reduction of administrative burden on citizens and businesses.

Unlike traditional approaches, where services are built around agency processes, here they are designed around data and user life events.

Key Feature of the Model

The proposed sequence is not strictly linear, but it establishes clear priorities. Attempting to implement later stages without first establishing foundational elements - such as data inventory and data governance - leads to increased complexity and reduced overall system effectiveness.
In other words, integration without data results in the scaling of uncertainty, while integration based on coherent data results in the scaling of efficiency.

Practical Implications (Policy Level)

Transitioning to a data-first model does not require abandoning existing integration solutions. On the contrary, existing infrastructure can become a valuable asset if embedded within a broader data governance framework.

In this context, several practical directions can be identified to gradually shift the focus of digital transformation toward a data-first approach.

First, establishing a national data catalog should be a priority. Such a catalog should describe not only existing information systems, but also the datasets themselves, including their content, structure, and responsible entities. This enables a transition from fragmented knowledge to a systematic understanding of the data landscape.

Second, it is essential to institutionalize data governance. This involves defining data ownership roles, identifying sources of truth, and implementing processes that ensure data quality and consistency. Without this, even the most advanced technological solutions will not produce sustainable outcomes.

Third, data standardization must be prioritized. Developing and implementing unified classifications, reference data, and data models is a critical condition for achieving semantic interoperability. It is at this level that the foundation is laid for consistent data use across systems and services.
Fourth, the development of integration infrastructure should be aligned with progress in data governance. This helps avoid situations where technical capabilities for data exchange outpace the readiness of data itself.

Finally, when designing digital services, it is important to consider not only existing processes but also data availability and quality. This approach supports the creation of simpler, end-to-end, user-centric services, reduces integration costs, accelerates implementation, and enables sustainable scaling.

Next Logical Step
As a practical continuation of the proposed approach, a logical next step is the launch of a national data catalog initiative.
Such an initiative can be implemented in stages, starting with priority data domains and gradually expanding across government institutions. A key success factor is the combination of methodological design, institutional support, and technical implementation.
As quick wins, the following actions can be undertaken:

- a pilot data catalog covering 3–5 key government agencies;
- the development of a basic set of classifications (e.g., addresses, organizations);
- the designation of data owners for a limited number of priority registries.

Limitations

This study has several limitations. First, it is based on qualitative analysis and does not include a quantitative assessment of digital transformation outcomes. Second, the primary empirical focus is on the case of the Kyrgyz Republic, which may limit the generalizability of the findings.
At the same time, the comparative analysis of Central Asian countries suggests that the identified patterns are representative of a broader regional context.

Conclusion

The experience of digital transformation in the public sector, including the case of Kyrgyzstan, demonstrates that technological integration alone is not sufficient to build an effective digital government.

Integration platforms, even when based on advanced technologies, can only enable the transfer of data. Their true value depends on how well that data is understood, aligned, and governed.

In this sense, the central challenge of digital transformation is not the development of data exchange technologies, but the establishment of a coherent data governance environment.

The transition to such an approach is not immediate and requires sequential institutional changes. At the same time, it creates the conditions for more sustainable digital services, improved government effectiveness, and the emergence of a truly data-driven state.

Integration thus becomes not the starting point of digital transformation, but its natural outcome.

This conclusion has direct implications for digital transformation strategies, shifting the focus from system integration to data governance as the foundational element of digital government.

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